

Dr. Doc:

Exam: Physiology

Year: 2014 E.C

Dr. Doc Tutorial

1. Wrong about shift of the Hb-oxygen dissociation curve of blood to the right

- A. Occurs in the pulmonary capillaries
- B. Occurs if blood temperature rises
- C. Favors oxygen delivery to the tissues
- D. Favors hydrogen binding to O₂ E. Increases per cent blood oxygen saturation

Answer: D. Favors hydrogen binding to O₂ ✗

Explanation: A rightward shift represents a decreased affinity of hemoglobin for oxygen, facilitating oxygen delivery (unloading) to active tissues. This shift is triggered by factors associated with high metabolism, such as increased temperature, increased (PCO₂), and increased H⁺ concentration.

Option D is wrong (the correct choice to select) because hydrogen ions bind to the amino acid chains on hemoglobin (the Bohr effect), not to the (O₂) molecules.

Memory: Right = Release O₂.

2. Wrong about Vital capacity is

- A. The volume of air expired from full inspiration to full expiration.
- B. Reduced as one grows older.

- C. Greater in men than in women of the same age and height
- D. Not related more to total body mass than to lean body mass
- E. The sum of the inspiratory and expiratory reserve volumes

Answer: E. The sum of the inspiratory and expiratory reserve volumes

Explanation: Vital Capacity (VC) is the maximum volume of air that can be voluntarily exhaled after a maximal inspiration.

It is mathematically defined as the sum of three distinct lung volumes: Tidal Volume (TV) + Inspiratory Reserve Volume (IRV) + Expiratory Reserve Volume (ERV).

Remember:

$$VC = IRV + TV + ERV$$

Residual volume is not included in vital capacity.

3. In pulmonary capillary blood

- A. Carbonic anhydrase in erythrocytes catalyses the formation of H^+ and HCO_3^-
- B. Hydrogen ions dissociate from hemoglobin.
- C. The fall in PO_2 is of greater magnitude than the fall in PCO_2
- D. The oxygen content is linearly related to alveolar PO_2
- E. The pH is higher than in blood in the pulmonary artery.

Answer: E. The pH is higher than in blood in the pulmonary artery.

Explanation: As blood flows through the pulmonary capillaries, carbon dioxide (CO_2) diffuses out of the blood and into the alveoli to be exhaled. This elimination of (CO_2) drives the bicarbonate buffer

equation in reverse (converting $\text{HCO}_3^- + \text{H}^+$ back into CO_2 and H_2O), which consumes free hydrogen ions (H^+). As a result, the H^+ concentration drops, making the pH of the pulmonary capillary blood higher (more alkaline) than the mixed venous blood arriving via the pulmonary artery.

In the lungs: CO_2 leaves $\rightarrow \text{H}^+$ decreases $\rightarrow \text{pH}$ increases.

So pulmonary capillary blood becomes more alkaline than incoming venous blood

4. Complete obstruction of a major bronchus usually results in

- A. Collapse of the alveoli supplied by the bronchus.
- B. A rise in local intrapleural pressure
- C. An increase in physiological dead space
- D. An increase in blood flow to the lung tissue supplied by the bronchus
- E. Hypoxemia.

Answer: A. Collapse of the alveoli supplied by the bronchus.

Explanation: Complete blockage of a bronchus prevents fresh air from entering the distal airway. The trapped air already inside the distal alveoli is rapidly absorbed into the local capillary blood, causing those alveoli to completely deflate and collapse—a pathological state known as absorption atelectasis.

5. Lack of surfactant substances as occur in infants with respiratory distress syndrome (RDS) results in:

- A. An increased lung compliance
- B. Stabilization of alveolar volume
- C. An increased retroactive force of the lungs by increasing alveolar surface tension
- D. A reduced alveolar-arterial O_2 tension difference
- E. A decrease in the filtration forces in the pulmonary capillaries

Answer: C. An increased retroactive force of the lungs by increasing alveolar surface tension

Explanation: Pulmonary surfactant reduces the surface tension at the air-water interface of the alveoli, preventing them from collapsing and making the lungs easy to expand. Without surfactant, alveolar surface tension rises drastically, which increases the elastic recoil (retroactive force) of the lungs, causing stiff lungs and widespread alveolar collapse.

6. The total lung capacity consists of:

- A. Vital capacity & tidal volume
- B. Residual volume & vital capacity
- C. Functional residual capacity & vital capacity
- D. Tidal volume & functional residual capacity
- E. Tidal volume & residual volume

Answer: B. Residual volume & vital capacity.

Explanation: Total Lung Capacity (TLC) represents the entire volume of air that the lungs can hold after a maximal inspiratory effort. It is the sum of the Vital Capacity (VC) (the maximum volume of air that can be voluntarily moved in and out of the lungs) and the Residual Volume (RV) (the remaining volume of air that cannot be exhaled even after maximal expiration).

7. The respiratory center responsible for the rhythm of respiration is located in the

- A. Ventral respiratory group
- B. Apneustic center
- C. Dorsal respiratory group
- D. Pneumotaxic center

E. Somewhere in the pons

Answer: C. Dorsal respiratory group.

Explanation: The Dorsal Respiratory Group (DRG), located in the medulla oblongata, contains inspiratory neurons that generate the basic rhythmic drive for quiet breathing by sending a repetitive "ramp" signal to the diaphragm.

Reminder: DRG → basic inspiratory rhythm.

The VRG becomes more important during forced breathing.

8. All of the following events occur during expiration except:

- A. Inspiratory muscles relax and the rib cage descends due to gravity
- B. Thoracic cavity volume decreases
- C. Intrapulmonary volume decreases
- D. Intrapulmonary pressure rises above atmospheric pressure (+1 mm Hg)
- E. Gases flow out of the lungs against the pressure gradient

Answer: E. Gases flow out of the lungs against the pressure gradient.

Explanation: During expiration, the relaxation of the inspiratory muscles and elastic recoil of the lung tissue decrease the thoracic volume, which causes intrapulmonary pressure to rise slightly above atmospheric pressure (to about +1 mmHg). This creates a gradient where air flows down (with) the pressure gradient from the higher pressure in the lungs to the lower pressure in the atmosphere, not against it.

9. Not the function of surfactant is:

- A. Increases surface area
- B. Decreases pressure in the alveoli
- C. Prevents lung collapse
- D. Increases work of breathing
- E. Facilitates gas exchange

Answer: D. Increases work of breathing

Explanation: Surfactant increases lung compliance, which decreases the work of breathing by making the lungs significantly easier to expand during inspiration.

10. The peripheral chemoreceptors are:

- A. Located on the ventral surface of the medulla oblongata
- B. More sensitive to CO₂
- C. Excited when PaO₂ is low
- D. Inactive during acidosis
- E. More sensitive to H⁺

Answer: C. Excited when PaO₂ is low.

Explanation: The peripheral chemoreceptors (located in the carotid and aortic bodies) are uniquely sensitive to arterial hypoxemia. They are strongly stimulated when arterial oxygen tension (PaO₂) falls below normal levels (typically below 60 mmHg). Conversely, the central chemoreceptors are located on the ventral surface of the medulla oblongata and are primarily responsive to changes in (CO₂) and pH.

11. Which structure of the following does not belong to the respiratory portion of the breathing organ?

- A. Terminal bronchioles
- B. Respiratory bronchioles
- C. Alveolar ducts
- D. Alveolar sacs
- E. None of the above

Answer: A. Terminal bronchioles.

Explanation: The respiratory system is split into a conducting zone and a respiratory zone. The terminal bronchioles are the final branches of the conducting zone; they do not contain alveoli and cannot participate in gas exchange. The respiratory portion begins downstream at the respiratory bronchioles, where alveoli first emerge.

12. Which one of the following is wrong about the normal oxygen tension?

- A. Venous blood = 40 mmHg
- B. Arterial blood = 95 mmHg
- C. Alveolar air = 104 mmHg
- D. Inspired air = 159 mmHg
- E. In the tissue > 40 mmHg.

Answer: E. In the tissue > 40 mmHg.

Explanation: Normal cellular metabolic pathways continuously consume oxygen, which keeps tissue (PO₂) low. At rest, tissue (PO₂) is equal to or less than 40 mmHg (typically around 23–40 mmHg). This

low tissue tension is essential because it establishes the steep pressure gradient needed for oxygen to diffuse from the arterial blood (approx. 95 mmHg) into the surrounding tissue cells.

13. Which structure of the following is responsible for respiratory rate and depth of breathing?

- A. Ventral respiratory group
- B. Dorsal respiratory group
- C. Apneustic center
- D. Pneumotaxic center
- E. Cerebral cortex

Answer: D. Pneumotaxic center.

Explanation: The pneumotaxic center, located in the upper pons, acts as an "off-switch" for the inspiratory ramp signal generated by the medulla. By limiting the duration of inspiration, it directly regulates the depth of breathing, which secondarily regulates the respiratory rate (a stronger pneumotaxic signal cuts inspiration short, leading to shallow, rapid breaths).

14. Possible mechanisms that would increase the rate of pleural fluid formation include:

- A. Decreased permeability of the pleural capillaries
- B. Decreased hydrostatic pressure in the pleural capillaries
- C. High colloid osmotic (oncotic) pressure in the pleural capillaries
- D. Obstruction of the pulmonary lymphatic channels
- E. None of the above

Answer: D. Obstruction of the pulmonary lymphatic channels.

Explanation: Pleural fluid is continuously produced by the pleural capillaries and drained out of the pleural cavity by the lymphatic system. Obstructing these lymphatic drainage pathways causes fluid to accumulate in the pleural space, leading to a pleural effusion.

15. Bronchial obstruction is likely to be relieved by

- A. Stimulation of cholinergic receptors
- B. Stimulation of beta adrenoceptors
- C. Histamine
- D. Stimulation of mast cell membranes
- E. Slow reactive substance of anaphylaxis

Answer: B. Stimulation of beta adrenoceptors.

Explanation: Bronchial smooth muscle contains beta-2 adrenergic receptors. Activating these receptors (via sympathetic stimulation or beta-agonist medications like albuterol) causes the smooth muscle to relax, resulting in bronchodilation and relief from obstruction. Cholinergic stimulation, histamine, and anaphylactic substances all cause bronchoconstriction instead.

16. The CO₂ dissociation curve for whole blood shows that

- A. Its shape is sigmoid.
- B. Blood saturates with CO₂ when PCO₂ exceeds normal alveolar levels.
- C. Blood contains some CO₂ even when the PCO₂ is zero.
- D. Oxygenation of the blood drives CO₂ out of the blood.
- E. Adding CO₂ to the blood drives O₂ out of the blood.

Answer: D. Oxygenation of the blood drives CO₂ out of the blood.

Explanation: This phenomenon is called the Haldane effect. In the lungs, when oxygen binds to hemoglobin, it alters the hemoglobin molecule's structure. This change reduces hemoglobin's affinity for carbon dioxide and hydrogen ions, helping to release bound (CO₂) into the alveoli so it can be exhaled.

17. Of the following, identify an important chemical buffer:

A. CO₂

B. O₂

C. H⁺

D. Hb

E. All

Answer: D. Hb.

Explanation: Hemoglobin (Hb) is a highly effective intracellular chemical buffer within red blood cells. It contains numerous histidine residues that can bind free hydrogen ions (H⁺) as oxygen is unloaded, which prevents drastic drops in blood pH during tissue transit.

18. Which of the following acts as an inward force in the pulmonary fluid dynamics?

A. Plasma hydrostatic pressure

B. Plasma oncotic pressure

C. Interstitial fluid oncotic pressure

D. Interstitial fluid hydrostatic pressure

E. B and D

Answer: E. B and D.

Explanation: In capillary fluid exchange, plasma oncotic pressure (pulling fluid back into the vessels via blood proteins) acts as a powerful inward force. Additionally, if interstitial fluid hydrostatic pressure is positive, it also acts as an inward physical force pushing fluid back into the capillary space.

19. Which one is true about VA/Q?

- A. Apical region of the lung is more perfused
- B. There is possible matching of VA and Q in the lungs
- C. VA/Q is smaller in the basal region of the lungs
- D. VA/Q is larger in the basal region of the lungs
- E. Whenever VA/Q is below normal, there is inadequate perfusion

Answer: C. VA/Q is smaller in the basal region of the lungs.

Explanation: Gravity causes both alveolar ventilation (VA) and capillary perfusion (Q) to increase toward the base of the upright lung. However, perfusion increases far more rapidly than ventilation toward the base. Because the increase in perfusion outpaces ventilation, the ventilation-perfusion ratio (VA/Q) is lower (smaller) at the base (about 0.6) and higher at the apex (about 3.0).

20. Decrease in pCO₂, decrease in H⁺ and increased PO₂ causes

- A. Hyperventilation
- B. Hypoventilation

- C. Hypercapnea
- D. Hypoxia
- E. Nothing happens

Answer: B. Hypoventilation.

Explanation: The primary chemical drive to breathe is monitored by chemoreceptors that detect rising (PCO_2) and rising H^+ levels, alongside falling (PO_2). When (PCO_2) drops, pH rises (fewer H^+ ions), and (PO_2) is high, the chemical stimulation to the respiratory centers is minimized, resulting in hypoventilation.

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21. Which of the following doesn't shift Hb-O₂ dissociation curve to the right down?

- A. An increase in PCO_2
- B. An increase in temperature
- C. An increase in the concentration of H^+
- D. An increase in 2,3-diphoglycerate
- E. An increase in pH

Answer: E. An increase in pH.

Explanation: An increase in pH represents a decrease in H^+ concentration (alkalosis). This increases hemoglobin's affinity for oxygen, shifting the dissociation curve to the left, which favors oxygen binding rather than unloading.

22. Not true of the COHb

- A. CO interferes with O₂ transport
- B. CO affects PO₂
- C. CO is undetectable
- D. CO shifts Hb-O₂ dissociation curve to the left
- E. CO has no physical signs of hypoxemia

Answer: B. CO affects PO₂.

Explanation: Carbon monoxide (CO) binds to the iron centers of hemoglobin, rendering them unavailable for oxygen transport. However, the partial pressure of oxygen (PO₂) measures only the physically dissolved oxygen in the blood plasma, which is unaffected by the presence of CO. Thus, CO does not affect dissolved (PO₂), making this statement false (and the correct answer).

23. The cranial nerves involved in the regulation of respiration are:

- A. Vagus and trigeminal
- B. Vagus and glossopharyngeal
- C. Vagus and facial
- D. Glossopharyngeal and hypoglossal
- E. Vagus and hypoglossal

Answer: B. Vagus and glossopharyngeal.

Explanation: The glossopharyngeal nerve (CN IX) carries sensory afferent signals from the carotid bodies, and the vagus nerve (CN X) carries sensory afferent signals from the aortic bodies and lung stretch receptors to the medullary respiratory center.

24. The major means by which CO₂ is transported is:

- A. Dissolved form
- B. Carbaminohemoglobin
- C. Bicarbonate
- D. Free form
- E. Bound with H⁺

Answer: C. Bicarbonate.

Explanation: Approximately 70% of the carbon dioxide produced by metabolism is transported in the blood in the form of dissolved bicarbonate ions (HCO₃⁻).

25. The apex of the upright human lung compared with the base has:

- A. A higher PCO₂
- B. A higher ventilation
- C. A lower pH in end-capillary blood
- D. A higher blood flow
- E. None of the above

Answer: E. None of the above.

Explanation: Due to the gravitational gradient in an upright lung, the apex has lower blood flow, lower ventilation, a lower (PCO₂) (because (VA/Q) is high), and a higher end-capillary pH. Since statements A, B, C, and D are all physiologically incorrect descriptions of the apex, E is the correct option.

26. Air-tight chest wall is important in that it:

- A. Increases lung compliance
- B. Decreases the work of breathing
- C. Creates negative pressure in the chest
- D. Prevents lung collapse
- E. Causes the lung to become "dry"

Answer: D. Prevents lung collapse.

Explanation: An airtight thoracic cavity is essential to maintain the subatmospheric (negative) intrapleural pressure that pulls outward on the lung tissue, opposing the lung's natural elastic inward recoil and preventing total collapse.

27. Of the following that is the respiratory function of the respiratory system is:

- A. Defense
- B. Phonation
- C. Gas conduction
- D. Olfaction
- E. Blood pressure regulation

Answer: C. Gas conduction.

Explanation: While defense, speaking, smelling, and blood pressure control are key non-respiratory functions, gas conduction is the primary respiratory function designed to transport fresh air to the gas-exchanging alveoli.

28. The same amount PO₂ taken up from the lungs, reaches the cell. The reason is:

- A. RBCs lack nucleus
- B. RBCs lack mitochondria
- C. Krebs's cycle occurs in the RBCs. D. RBCs lack Hb
- E. None of the above

Answer: B. RBCs lack mitochondria.

Explanation: Mature red blood cells (RBCs) lack mitochondria and generate energy strictly via anaerobic glycolysis. Because they do not perform aerobic cellular respiration, they do not consume any of the oxygen they transport from the lungs, ensuring that 100% of the bound oxygen reaches systemic tissues.

29. When respiratory drive for increased pulmonary ventilation becomes greater than normal, a special set of respiratory neurons that are inactive during normal quiet breathing then becomes active... These neurons are located in which of the following structures?

- A. Apneustic center
- B. Dorsal respiratory group
- C. Nucleus of the tractus solitarius D. Pneumotaxic center
- E. Ventral respiratory group

Answer: E. Ventral respiratory group.

Explanation: During normal quiet breathing, inspiration is an active process driven by the Dorsal Respiratory Group (DRG), while expiration is passive. When ventilation demands rise (e.g., during

exercise), the Ventral Respiratory Group (VRG) is recruited to actively drive both forced inspiration and forced expiration.

30. Which one of the following is true about tidal volume

- A. The amount of air that normally moves into (or out of) the lung with each respiration.
- B. The amount of air that enters the lung but does not participate in gas exchange.
- C. The largest amount of air expired after maximal expiratory effort.
- D. The largest amount of gas that can be moved into and out of the lungs in one minute.
- E. The volume of air one breathes in with exertion

Answer: A. The amount of air that normally moves into (or out of) the lung with each respiration.

Explanation: Tidal Volume (TV) is defined as the volume of air inspired or expired with each single, normal, resting

31. O₂ delivery to the tissues would be reduced to the greatest extent in

- A. A normal subject breathing 100% O₂ on top of Mt. Everest.
- B. A normal subject running a marathon at sea level.
- C. A patient with carbon monoxide poisoning.
- D. A patient who has ingested cyanide.
- E. A patient with moderately severe metabolic acidosis

Answer: C. A patient with carbon monoxide poisoning.

Explanation: Carbon monoxide (CO) poisoning is devastating because CO binds to hemoglobin with an affinity up to 250 times greater than oxygen, occupying binding sites and shifting the remaining curve to the left (preventing oxygen unloading at tissues). Cyanide poisoning blocks cellular oxygen utilization at the mitochondrial level, but arterial oxygen delivery (the transport of oxygen in the blood) remains high.

32. Which of the following hormone is secreted in excess when a person climbing a mountain?

- A. Anti Natriuretic Factor
- B. Adrenaline
- C. Erythropoietin
- D. Thyroxine
- E. Aldestrone

Answer: C. Erythropoietin.

Explanation: The low atmospheric pressure at high altitudes reduces arterial (PO_2), triggering systemic hypoxia. This hypoxia is detected by the kidneys, which respond by secreting Erythropoietin (EPO). EPO stimulates the bone marrow to increase red blood cell production, expanding the blood's overall oxygen-carrying capacity over time.

33. Decreasing the surface area of the respiratory membrane would affect

- A. Inhalation
- B. Cellular respiration
- C. External respiration
- D. Mucus production
- E. Internal respiration

Answer: C. External respiration.

Explanation: External respiration is the gas exchange occurring between the air in the alveoli and the blood flowing in the pulmonary capillaries. Fick's Law of Diffusion dictates that reducing the surface area of the respiratory membrane (as seen in emphysema) directly decreases the rate of this gas exchange

34. In the lungs, surfactant substance is produced by

- A. Pneumocyte type I
- B. Pneumocyte type II
- C. Pneumocyte type III
- D. Alveolar macrophages
- E. White blood cell

Answer: B. Pneumocyte type II.

Explanation: Type II alveolar epithelial cells (Type II pneumocytes) are specialized cuboidal cells in the alveolar wall responsible for synthesizing, storing, and secreting surfactant.

35. The extent to which the lung expands is its

- A. Resistance
- B. Compliance
- C. Viscosity
- D. Recoiling ability
- E. Dryness

Answer: B. Compliance.

Explanation: Lung compliance measures the expandability or stretchability of the lungs. It is calculated as the change in lung volume divided by the change in transpulmonary pressure.

36. If oxygen content is 14 ml O₂/dL blood and oxygen capacity is 20 ml O₂/dL blood, [the saturation is]:

A. 80%

B. 60%

C. 70%

D. 65%

E. 75%

Answer: C. 70%

Explanation: Hemoglobin oxygen saturation (S-O₂) represents the percentage of hemoglobin binding sites that are occupied by oxygen, calculated as: $\text{Oxygen Saturation} = \text{Oxygen Content} \div \text{Oxygen Capacity} \times 100\%$

$$14\text{ml/dL} \div 20\text{ml/dL} \times 100\% = 70\%$$

Exam: Physiology

Year: 2015 E.C

Dr. Doc Tutorial

1. In a person running 10,000 m race which muscles contract during expiration?

- A) Diaphragm and external intercostal muscles
- B) Internal intercostal muscles and abdominal recti
- C) Sternocleidomastoid muscles
- D) Diaphragm and internal intercostal muscles
- E) None of the above

Correct Answer: B. Internal intercostal muscles and abdominal recti

Explanation: Quiet expiration during rest is entirely passive and relies on the elastic recoil of the lungs. However, during heavy exercise (like running a 10,000 m race), ventilation requirements increase dramatically and expiration becomes active (forced expiration). This requires the active contraction of expiratory muscles, primarily the abdominal muscles (rectus abdominis, obliques) and the internal intercostal muscles. The diaphragm and external intercostals are primary muscles of inspiration, and the sternocleidomastoid is an accessory muscle of inspiration.

2. The normal pleural pressure of an individual is approximately -5 cm H₂O during resting conditions immediately before inspiration. What is the pleural pressure (in cm H₂O) during inspiration?

- A) -1.5
- B) +1
- C) 0
- D) -7.5
- E) None of the above

Correct Answer: D. -7.5

Explanation: Intrapleural (pleural) pressure is the slightly negative pressure in the thin fluid space between the lung visceral pleura and the chest wall parietal pleura. At the beginning of inspiration, it is about -5 cm H₂O (representing the balance of the chest wall pulling outward and the lung elastic recoil pulling inward). During inspiration, chest wall expansion increases the thoracic volume, which intensifies the suction force, causing the intrapleural pressure to become more negative, averaging about -7.5 cm H₂O at the end of a normal quiet inspiration.

3. Compliance of the lung:

- A) is about 500 ml/cm H₂O
- B) is decreased in emphysematous changes
- C) is higher in small children than in adults
- D) is increased in preterm infants E) None of the above

Correct Answer: E. None of the above

Explanation: Let's evaluate each option to see why they are incorrect: Option A is incorrect: The total compliance of both lungs in a normal adult averages about 200 mL/cm H₂O (or 0.2 L/cm H₂O), not 500 mL. Option B is incorrect: Lung compliance is actually increased (not decreased) in pulmonary emphysema. This is because emphysematous changes destroy the alveolar septa and elastic fibers, reducing the lung's elastic recoil, making it much easier to distend. Option C is incorrect: Absolute lung compliance is lower (not higher) in small children than in adults because of their much smaller lung volumes (compliance is volume-dependent). Option D is incorrect: Compliance is severely decreased (not increased) in preterm infants because of a deficiency of surfactant. This deficiency leads to very high alveolar surface tension, making the lungs stiff and prone to collapse. Thus, E is the correct choice.

4. Surfactant:

- A) is formed by pulmonary capillaries endothelial cells
- B) increases alveolar fluid surface tension

- C) can be deficient in Premature infants
- D) is composed of protein only
- E) None of the above

Correct Answer: C. can be deficient in Premature infants

Explanation: Surfactant is a complex mixture of phospholipids (predominantly dipalmitoyl phosphatidylcholine/lecithin) and surfactant-associated proteins (not protein only). It is synthesized and secreted by Type II pneumocytes (not capillary endothelial cells) inside the alveoli. Its primary function is to dramatically decrease (not increase) the surface tension of alveolar fluid. Surfactant production only begins in the latter stages of gestation; therefore, premature infants can be highly deficient in surfactant, causing them to suffer from Infant Respiratory Distress Syndrome (IRDS).

5. Which of the following lung volumes or capacities can be measured by direct spirometry?

- A) FRC
- B) RV
- C) TLC
- D) VC
- E) None of the above

Correct Answer: D. VC

Explanation: Spirometry can only measure volumes of air that can be inspired or expired from the lungs. Because the lungs cannot be completely emptied even after a maximal, forceful expiration, the Residual Volume (RV) cannot be measured directly by simple spirometry. Consequently, any lung capacity that includes RV cannot be measured directly either. This includes Functional Residual Capacity (FRC) (FRC =

ERV + RV) and Total Lung Capacity (TLC) ($TLC = VC + RV$). Vital Capacity (VC), which is the maximum amount of air a person can expire after a maximum inspiration ($VC = IRV + VT + ERV$), does not include the residual volume and therefore can be measured directly.

6. In an individual who has a vital capacity (VC) of 5 L, a tidal volume (VT) of 0.5 L, an inspiratory capacity of 3.5 L, and a functional residual capacity (FRC) of 2.5 L. What is his expiratory reserve volume (ERV)?

- A) 2.0 L
- B) 1.5 L
- C) 1.0 L
- D) 2.5 L
- E) None of the above

Correct Answer: B. 1.5 L

Explanation: Based on standard spirometric definitions, Vital Capacity (VC) is the sum of Inspiratory Capacity (IC) and Expiratory Reserve Volume (ERV). Mathematically: $VC = IC + ERV$. Rearranging this formula to solve for the unknown ERV gives: $ERV = VC - IC = 5.0\text{ L} - 3.5\text{ L} = 1.5\text{ L}$. Note that the other values provided ($VT = 0.5\text{ L}$ and $FRC = 2.5\text{ L}$) are extra information and are not required to solve this specific equation.

7. A capacity that remains in the lungs after a tidal volume (VT) is expired?

- A) FRC
- B) VC
- C) TLC
- D) IC

E) None of the above

Correct Answer: A. FRC

Explanation: Functional Residual Capacity (FRC) is defined as the volume of air remaining in the lungs at the end of a normal tidal expiration. It represents the resting equilibrium volume of the respiratory system and is equal to the sum of Expiratory Reserve Volume (ERV) and Residual Volume (RV): $FRC = ERV + RV$.

8. A 59-years-old man has a pulmonary embolism that completely blocked blood flow to his left lung. As a result, which of the following will occur?

A) Ventilation/perfusion (VA/Q) ratio in the left lung will be zero

B) Systemic arterial PO_2

will be elevated

C) Alveolar PO_2 air

in the left lung will be approximately equal to the PO_2

D) VA/Q ratio in the left lung will be lower than in the right lung

E) None of the above Correct

Answer: C. Alveolar PO_2

in inspired in the left lung will be approximately equal to the PO_2 in inspired air

Explanation: A complete pulmonary vascular blockage means that blood flow (perfusion, Q) to the left lung is zero. Because ventilation (VA) continues but perfusion is zero, the ventilation-perfusion ratio (VA/Q) is infinity (physiological dead space), making Option A and D incorrect. Under these conditions, there is no gas exchange in the left lung, so the alveolar gas cannot transfer oxygen into the blood or

receive carbon dioxide from it. Consequently, the alveolar air of the left lung equilibrates perfectly with humidified inspired air, raising the left alveolar PO_2

to about 149 mm Hg (compared to the normal 104 mm Hg) and reducing the left alveolar PCO_2 to near-zero (0.3 mm Hg).

9. An increase in physiological dead space would be expected with:

- A) Pulmonary embolism that blocks pulmonary blood flow
- B) Atelectasis or collapse of one lung
- C) Bronchoconstriction
- D) B and C
- E) None of the above

Correct Answer: A. Pulmonary embolism that blocks pulmonary blood flow

Explanation: Physiological dead space is the sum of anatomical dead space (conducting airways) and alveolar dead space (ventilated alveoli that receive no blood flow). A pulmonary embolism blocks capillary perfusion to a region of the lungs. The affected alveoli remain ventilated, but since blood flow is absent, they are incapable of gas exchange. This creates a large volume of alveolar dead space, increasing the overall physiological dead space. In contrast, atelectasis (alveolar collapse) represents a loss of ventilation but continued blood flow ($VA/Q = 0$), which acts as a physiological shunt rather than dead space.

10. As we go from the top of the erect lung to the bottom:

- A) VA/Q is higher at base than apex
- B) Ventilation is higher at base than apex
- C) Alveolar PCO_2 at apex is higher than at the base
- D) Alveolar PO_2 at base is higher than at the apex

E) None of the above

Correct Answer: B. Ventilation is higher at base than apex

Explanation: In an upright (erect) individual, gravity pulls the blood and the lung tissue downward, creating hydrostatic pressure gradients. As a result, both ventilation (VA) and perfusion (Q) increase linearly from the top (apex) to the bottom (base) of the lung, meaning ventilation is higher at the base (Option B is correct). However, perfusion increases much more rapidly than ventilation, causing the ventilation-perfusion (VA/Q) ratio to be significantly higher at the apex (approx. 2.5Ö ideal) and lower at the base (approx. 0.6Ö ideal). Due to the high VA/Q ratio at the apex, oxygen loading is maximized and carbon dioxide unloading is high, meaning the apex has a higher alveolar PO₂

and a lower alveolar PCO₂ than the base.

11. Pulmonary circulation is:

A) high-pressure circulation

B) high-flow circulation

C) low-flow circulation

D) to supply systemic arterial blood to the trachea, bronchial tree, and supporting tissues of the lung

E) None of the above

Correct Answer: B. high-flow circulation Physiological

Explanation: The pulmonary circulation is a low-pressure, high-flow circuit. It receives 100% of the cardiac output (high flow) but operates under low arterial pressure (systolic/diastolic of $\approx 25/8$ mm Hg). The systemic circuit, by comparison, is a high-pressure, high-flow circuit. The bronchial circulation is a systemic vascular subdivision that represents a high-pressure, low-flow circuit (1–2% of cardiac output) supplying oxygenated blood to the trachea, bronchial tree, and supporting lung tissues.

12. In which vascular bed does hypoxia cause vasoconstriction?

- A) Coronary
- B) Pulmonary
- C) Cerebral
- D) Muscle
- E) None of the above

Correct Answer: B. Pulmonary

Explanation: In the pulmonary vasculature, localized alveolar hypoxia (alveolar PO₂

< 73 mm Hg) triggers hypoxic pulmonary vasoconstriction. This is a critical protective

local reflex that increases vascular resistance in poorly ventilated regions, shunting blood toward well-ventilated alveoli to preserve VA/Q matching. This reaction is the exact opposite of systemic vascular beds (like the coronary, cerebral, and skeletal muscle circulations), which dilate in response to hypoxia to maximize perfusion and oxygen delivery to hypoxic tissues.

13. In an alveoli with a normal ventilation perfusion ratio, blood at the end of the pulmonary capillary:

- A) Has a PO₂ similar to blood in the left ventricle
 - B) Has a PO₂ similar to that in the left atrium
 - C) Has a PO₂ higher than blood from the aorta
 - D) Has a percent saturation of 95%
 - E) None of the above
- Correct

Answer: C. Has a PO₂

higher than blood from the aorta

Explanation: At the end of a normal pulmonary capillary, oxygen has fully equilibrated with the alveolar air, achieving a PO₂

of about 104 mm Hg and a hemoglobin saturation

of 97%. However, as this blood returns to the left atrium and left ventricle, it is mixed with a small amount of deoxygenated shunt blood (such as bronchial venous drainage, which is about 2% of total cardiac output, arising from the bronchial venous drainage and thebesian veins). This shunt dilutes the oxygen content, lowering the systemic arterial PO₂

in the aorta, left atrium, and

left ventricle to about 95 mm Hg (97% saturation). Thus, end-capillary blood has a higher PO₂ (104 mm Hg) than aortic, left atrial, or left ventricular blood (95 mm Hg).

14. The forces governing the diffusion of a gas through respiratory membrane include the pressure difference across the membrane (ΔP), the cross-sectional area of the membrane (A), the solubility of the gas (S), the distance of diffusion (d), and the molecular weight of the gas (MW). Which changes increase the diffusion of a gas through a biological membrane?

- A) Increase ΔP , Increase A , Increase S , Increase d , Increase MW .
- B) Increase ΔP , Increase A , Increase S , Increase d , Decrease MW .
- C) Increase ΔP , Decrease A , Increase S , Decrease d , Decrease MW .
- D) Increase ΔP , Increase A , Increase S , Decrease d , Decrease MW .
- E) None of the above

Correct Answer: D. Increase ΔP , Increase A , Increase S , Decrease d , Decrease MW .

Explanation: According to Fick's Law of Diffusion, the net rate of gas diffusion

across a biological membrane is expressed by the equation: Diffusion Rate $\propto \Delta P \cdot A \cdot S \cdot d^{-1} \cdot MW^{-1}$

Therefore, to increase the rate of diffusion, we must increase the factors in the numerator (increase pressure gradient ΔP , surface area A , and gas solubility S) and decrease the factors in the denominator (decrease membrane thickness/distance d and molecular weight MW). This matches Option D.

15. Which one is among factors that increase respiratory membrane thickness thus impairs gas exchange

- A) removal of an entire lung
- B) emphysema
- C) edema fluid in the interstitial space of the membrane and in the alveoli
- D) Atelectasis or collapse of one lung
- E) None of the above

Correct Answer: C. edema fluid in the interstitial space of the membrane and in the alveoli

Explanation: The average thickness of the respiratory membrane is only 0.6 μm . In pulmonary edema, fluid leaks out of the capillaries and accumulates within the interstitial space and the alveoli. This fluid layer physically increases the distance of diffusion (d), thickening the barrier and severely impairing the rate of gas exchange. In contrast, the removal of a lung or chronic emphysema leads to a reduction in the total surface area (A) of the membrane, rather than thickening the individual membrane layers.

Question 16 Ninety seven percent of oxygen is carried in the blood in the form of A) dissolved oxygen in blood plasma B) oxyhemoglobin C) carbaminohemoglobin D) in the form of bicarbonate ion E) None of the above

Correct Answer: B. oxyhemoglobin Physiological Explanation: Oxygen is transported by the blood in two forms: (1) physically dissolved in the plasma, which accounts for only 3% of transport (approximately 0.17 mL O₂/dL blood delivered to tissues), and (2) chemically bound to the heme groups of hemoglobin inside erythrocytes, forming oxyhemoglobin, which accounts for the remaining 97% (delivering approximately 5 mL O₂/dL blood at rest).

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- D) in the form of bicarbonate ion
- E) None of the above

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17. Oxygen dissociation curve shift to the right:

- A) Due to decrease in blood H⁺ concentration
- B) Means the O₂ affinity of hemoglobin is reduced
- C) Due to decrease in the concentration of 2,3-bisphosphoglycerate in the red cells
- D) Due to decrease in blood CO₂
- E) None of the above

Correct Answer: B. Means the O₂ affinity of hemoglobin is reduced

Explanation: A rightward shift of the oxygen-hemoglobin dissociation curve indicates that the oxygen affinity of hemoglobin is decreased, facilitating the unloading and release of oxygen to the tissues. A rightward shift is caused by factors that occur in metabolically active, demanding tissues: increased H⁺ (acidosis/decreased pH), increased PCO₂ (hypercapnia),

increased temperature, and increased 2,3-diphosphoglycerate (2,3-DPG) in the RBCs. Decreases in these variables shift the curve to the left.

18. Which one is incorrect

A) Carbon monoxide molecules displace O₂ from the hemoglobin

B) Carbon monoxide has about 250 times the affinity of O₂ for hemoglobin

C) In CO poisoning the PO₂
of the blood may be normal

D) In CO poisoning increased respiration rate in response to lack of O₂ usually reflected is absent

E) None of the above

Correct Answer: B. Carbon monoxide has about 250 times the affinity of O₂ for hemoglobin

Explanation: Let's analyze why Option B is incorrect (and therefore the correct answer for this 'incorrect' query): Option B is incorrect: The slides explicitly state that the binding affinity of hemoglobin for carbon monoxide (CO) is about 210 times (not 250 times) its affinity for oxygen. Option A is correct: CO binds competitively to the same iron-binding sites on the heme groups, directly displacing O₂. Option C is correct: Arterial PO₂

measures only

physically dissolved oxygen in the plasma, which is unaffected by CO binding to hemoglobin. Thus, arterial PO₂

is normal. Option D is correct: Because arterial PO₂ chemoreceptors (which respond to low PO₂ is normal, the peripheral , not oxygen content) are not stimulated. Therefore,

the hyperp- nea/ventilatory feedback response to tissue hypoxia is absent, making CO poisoning highly insidious.

19. Transport of carbon dioxide by blood is mainly in the form of

- A) bicarbonate ions
- B) carbaminohemoglobin
- C) dissolved carbon dioxide in blood plasma
- D) oxyhemoglobin
- E) None of the above

Correct Answer: A. bicarbonate ions

Explanation: Carbon dioxide is transported in blood in three forms: (1) dissolved in plasma (7%), (2) chemically bound as carbaminohemoglobin (23%), and (3) in the form of bicarbonate ions (HCO^-) (70%). Bicarbonate is the major transport mechanism.

20. Haldane effect:

A) is a shift of the oxygen-hemoglobin dissociation curve to the right in response to increases in blood CO₂

B) is a shift of the oxygen-hemoglobin dissociation curve to the right in response to increases in blood hydrogen ions

C) has a significant effect by enhancing the release of O₂ from the blood in the tissues

D) is the effect of binding of O₂ with hemoglobin that tends to displace CO₂ from the blood

E) None of the above

Correct Answer: D. is the effect of binding of O₂ with hemoglobin that tends to displace CO₂ from the blood

Explanation: The Haldane effect states that the binding of oxygen to hemoglobin in the lungs directly decreases hemoglobin's affinity for CO₂ and H⁺, promoting the displacement and unloading of carbon dioxide into the alveoli for expiration. (Conversely, deoxygenation of hemoglobin in the tissues increases its affinity for CO₂ and H⁺, facilitating CO₂ pickup). Note that Options A and B describe the Bohr effect. Option C is also a feature of the Bohr effect, not the Haldane effect.

21. Inspiration is initiated by discharge from:

A) The dorsal inspiratory group of neurons

B) The ventral group of neurons

C) Pneumotaxic center

D) A respiratory center in pons

E) None of the above

Correct Answer: A. The dorsal inspiratory group of neurons

Explanation: The primary rhythmic drive and basic pattern of breathing are controlled by the Dorsal Respiratory Group (DRG) of neurons, located in the nucleus of the tractus solitarius (NTS) in the medulla oblongata. The DRG neurons act as I-neurons (inspiratory), emitting repetitive, rhythmic action

potentials in a ramp-like manner (steadily increasing for 2 seconds) to stimulate primary inspiratory muscles like the diaphragm before abruptly stopping for 3 seconds to allow passive expiration.

22. The exchange of respiratory gasses (O₂ and CO₂) between alveoli and pulmonary capillaries

- A) Internal respiration
- B) Cellular respiration
- C) External respiration
- D) Pulmonary ventilation
- E) Breathing

Correct Answer: C. External respiration

Explanation: External respiration is specifically defined as the physiological process of gas exchange (oxygen loading and carbon dioxide unloading) across the respiratory membrane between the air in the alveoli and the blood flowing in the pulmonary capillaries. Internal respiration or cellular respiration refers to the exchange and metabolic utilization of these gases at the systemic tissue and cellular level. Pulmonary ventilation refers to the physical mechanical movement of air between the atmosphere and the lungs.

23. At the end of both inspiration and expiration, the intrapleural pressure

- A) Greater than atmospheric pressure
- B) Equal to atmospheric pressure
- C) Equal to alveolar pressure
- D) Greater than alveolar pressure
- E) None of the above

Correct Answer: E. None of the above

Explanation: At the static endpoints of both inspiration and expiration, airflow stops because the alveolar (intrapulmonary) pressure has equilibrated to become exactly equal to atmospheric pressure (0 cm H₂O). However, due to the continuous, opposing elastic recoil forces of the lungs and chest wall, the intrapleural pressure remains negative under all normal conditions (approximately -7.5 cm H₂O at the end of inspiration and -5 cm H₂O at the end of expiration). Thus, it is always lower than both atmospheric and alveolar pressure, making E the correct choice.

24. The rate of diffusion of a gas across the respiratory membrane decreases as the

- A) Respiratory membrane becomes thicker.
- B) Surface area of the respiratory membrane decreases.
- C) Molecular weight of the gas increases.
- D) Diffusion coefficient of the gas decreases.
- E) All are answers.

Correct Answer: E. All are answers.

Explanation: According to Fick's Law of Diffusion, the rate of gas transfer across the blood-gas barrier is inversely proportional to membrane thickness (d) and the molecular weight of the gas (MW), and directly proportional to the surface area (A) and the solubility/diffusion coefficient of the gas. Therefore, the diffusion rate decreases as:

thickness increases (A),

surface area decreases (B), molecular weight increases (C), or diffusion coefficient decreases (D).

Thus, all statements are correct.

25. During quiet breathing; around 70% of the pulmonary ventilation is mainly by which of the following muscles contraction?

- A) External intercostals
- B) Diaphragm and internal intercostals
- C) Diaphragm and external intercostals
- D) Internal intercostals and abdominal recti
- E) Diaphragm

Correct Answer: E. Diaphragm

Explanation: Normal, quiet resting breathing is driven almost entirely by the rhythmic contraction of the diaphragm (contributing roughly 70% of the total ventilation), which moves downward to lengthen the thoracic cavity. When the diaphragm contracts, it flattens and descends, increasing the vertical and transverse dimensions of the thoracic cavity, creating a sub-atmospheric pressure in the alveoli to draw air in. The external intercostals assist, but the diaphragm is the major contributor.

26. During inspiration, the intrapulmonary pressure

- A) Less than the pleural pressure by 1 mm Hg
- B) Greater than the atmospheric pressure by 3 mm Hg
- C) Less than the atmospheric pressure by 6 mm Hg
- D) Greater than the pleural pressure by 4 mm Hg
- E) Less than the pleural pressure by 3 mm Hg

Correct Answer: D. Greater than the pleural pressure by 4 mm Hg

Explanation: Let's calculate the pressure difference during inspiration: intrapulmonary (alveolar) pressure drops to a sub-atmospheric level of about -1 cm H₂O (or -0.74 mm Hg). Concurrently, pleural pressure drops to a more negative level of about -7.5 cm H₂O (or -5.5 mm Hg). The transpulmonary pressure ($P_{trans} = P_{alveolar} - P_{pleural}$) represents the force keeping the alveoli open: $[-1 \text{ cm H}_2\text{O}] - [-7.5 \text{ cm H}_2\text{O}] = +6.5 \text{ cm H}_2\text{O}$ (which is approximately 4.8 mm Hg). Thus, the intrapulmonary pressure is greater than the pleural pressure by approximately 4 mm Hg (or 6.5 cm H₂O), matching Option D.

27. When blood passes through the pulmonary capillaries, the hemoglobin increases the release of carbon dioxide because of

- A) Bohr effect
- B) Haldane effect
- C) Chloride shift
- D) Boyle's effect
- E) Dalton effect

Correct Answer: B. Haldane effect

Explanation: As venous blood passes through the pulmonary capillaries, it is exposed to high alveolar oxygen tension, causing oxygen to bind to hemoglobin. The oxygenation of hemoglobin causes it to become a stronger acid, which (1) directly releases hydrogen ions (H^+) that combine with bicarbonate to form carbonic acid and subsequently CO₂, and (2) structurally decreases its affinity for CO₂ bound to its terminal amine groups (carbaminohemoglobin). This direct displacement of carbon dioxide by oxygen binding is known as the Haldane effect.

28. Partial pressure of carbon dioxide in the

- A) Arterial end of tissue capillaries is greater than in the interstitial fluid.
- B) Intracellular is less than in the interstitial fluid.
- C) Venous end of pulmonary capillaries is less than in the alveoli.

D) Alveoli is greater than in the arterial end of pulmonary capillaries.

E) Venous ends of tissue capillaries is equal to in the interstitial fluid

Correct Answer: E. Venous ends of tissue capillaries is equal to in the inter- stitial fluid

Explanation: As blood flows through the systemic tissue capillaries, carbon dioxide diffuses down its steep partial pressure gradient from the high-pressure intracellular compartment (> 46 mm Hg) into the interstitial fluid (45 mm Hg), and then into the capillary blood (40 mm Hg at the arterial end). By the time blood reaches the venous end of the tissue capillaries, gas exchange has reached dynamic equilibrium, making the venous end PCO₂

equal to the PCO₂ in the interstitial fluid (45 mm Hg).

29. Which respiratory group "switch off" inspiratory signals, thus controlling the du- ration of filling phase of the lung cycle

A) Apneustic center

B) Dorsal respiratory group

C) Nucleus of the tractus solitarius

D) Pneumotaxic center

E) Ventral respiratory group

Correct Answer: D. Pneumotaxic center

Explanation: The pneumotaxic center , located in the nucleus parabrachialis of the upper pons, controls the duration of the lung filling phase. It does this by sending inhibitory neural signals to the Dorsal Respiratory Group (DRG) to control the "switch-off" point of the inspiratory ramp signal. Strong pneumotaxic signaling cuts off inspiration early (shortening the filling phase), which secondarily shortens expiration and increases the overall breathing frequency (up to 30–40 breaths/min). Weak signaling prolongs inspiration and slows down the breathing rate.

30. Which of the following best describes the effect of Hering-Breuer reflex

- A) Inhibit the respiratory center neurons and cause expiration
- B) Regulating the basic rhythm of breathing for infants
- C) Limits the duration of inspiration and prevents over inflation of the lungs
- D) Important only when the tidal volume is large during heavy exercise
- E) All are answers

Correct Answer: E. All are answers

Explanation: The Hering-Breuer reflex is a protective inflation-limiting reflex mediated by stretch receptors in the muscular walls of the bronchi and bronchioles. When activated by extreme lung distension (typically when tidal volume exceeds 1.5 L, such as during heavy exercise), it transmits inhibitory signals via the vagus nerve to the DRG. This arrests inspiration and initiates expiration (A), limits the duration of inspiration to prevent lung over-inflation (C), and is a crucial regulator of normal respiratory rhythm in infants (B). Therefore, all options are correct.

31. The effect of pH on the oxygen–hemoglobin dissociation curve is described by

- A) Bohr effect
- B) Haldane effect
- C) chloride shift
- D) Boyle's effect
- E) Dalton effect

Correct Answer: A. Bohr effect

Explanation: The Bohr effect is the physiological phenomenon where an increase in blood hydrogen ion concentration (decreased pH/acidosis) and carbon dioxide reduces the oxygen-binding affinity of hemoglobin, shifting the oxygen-hemoglobin dissociation curve to the right and downward. This directly enhances the unloading of oxygen in acidic, highly active metabolic tissues.

32. Which of the following correctly describes the oxygen–hemoglobin dissociation curve

- A) Shifts to the right in the pulmonary capillaries and to the left in the tissue capillaries.
- B) Shifts to the left in the pulmonary capillaries and to the right in the tissue capillaries.
- C) An increase PH, decrease CO₂ and decrease in temperature increase the ability of hemoglobin to release of O₂
- D) A decrease in PH, increase in CO₂ and increase in temperature decrease the ability of hemoglobin to release of O₂
- E) All are answers

Correct Answer: B. Shifts to the left in the pulmonary capillaries and to the right in the tissue capillaries.

Explanation: Let's evaluate the physiological shifts of the curve: In pulmonary capillaries: Carbon dioxide is expired, raising blood pH. This shifts the curve to the left, increasing hemoglobin's oxygen affinity to maximize oxygen loading in the lungs. In tissue capillaries: Metabolizing cells release CO₂ and acids, lowering blood pH. This shifts the curve to the right, lowering oxygen affinity to facilitate oxygen unloading to the tissues. (Thus, Option B is correct and Option A is incorrect). Option C and D are incorrect: An increase in pH (alkalosis) shifts the curve left, which increases oxygen affinity and therefore decreases (not increases) hemoglobin's ability to release oxygen.

33. Which of the following is correct during gas exchanges in the tissue at a time when hydrogen ions bind to hemoglobin

- A) Ability of Hemoglobin to releases oxygen is increase
- B) Transport of carbon dioxide increases
- C) pH inside the red blood cells does not decrease
- D) Affinity of hemoglobin for oxygen decreases
- E) All are answers

Correct Answer: E. All are answers

Explanation: In systemic tissue capillaries, carbon dioxide enters the RBC and is hydrated to form H^+ and HCO_3^- . The liberated H^+ ions bind directly to hemoglobin. This process triggers several crucial physiological mechanisms: 1. Acid-Base Buffering: By binding to the H^+ ions, hemoglobin acts as a powerful buffer, ensuring that the pH inside the red blood cell does not drop precipitously (C is correct). 2. Oxygen Unloading: The binding of H^+ structurally shifts hemoglobin's conformation, decreasing its oxygen affinity (D is correct), which in turn increases its ability to release oxygen to the tissues (A is correct). 3. Carbon Dioxide Transport: The deoxygenated hemoglobin has a higher affinity to form carbaminohemoglobin and carry CO_2 back to the lungs (the Haldane effect, B is correct). Thus, all options are correct.

34. The volume of air that actually used for exchanged between the atmosphere and the alveoli is

- A) 1100 mL
- B) 150 mL
- C) 500 mL
- D) 350 mL
- E) 1200 mL

Correct Answer: D. 350 mL

Explanation: With each normal breath, a tidal volume (V_T) of 500 mL is inspired. However, about 150 mL of this air remains trapped inside the conducting zone of the airways (anatomical dead space: nose, pharynx, trachea, bronchi), where no gas exchange capillaries are present. Thus, the volume of fresh atmospheric air that actually reaches the respiratory zone (alveoli, alveolar ducts) to participate in gas exchange is: $V_{\text{Alveolar}} = V_T - V_D = 500 \text{ mL} - 150 \text{ mL} = 350 \text{ mL}$

35. During pregnancy; pulmonary ventilation mainly occurred due to the contraction of

- A) External intercostals
- B) Diaphragm and internal intercostals
- C) Diaphragm and external intercostals
- D) Internal intercostals and abdominal recti
- E) Diaphragm

Correct Answer: A. External intercostals

Explanation: In a pregnant individual, the expanding gravid uterus physically pushes the abdominal contents and the diaphragm upward into the thoracic cavity, mechanically restricting the diaphragm's downward excursion. To maintain normal tidal volumes and compensate for this physical restriction, the individual relies much more heavily on thoracic/costal chest expansion, which is driven by the active contraction of the external intercostal muscles.

36. The volume of air exchanged between the atmosphere and alveoli per minute

- A) Minute Ventilation
- B) respiratory rate

C) Pulmonary Ventilation

D) Alveolar Ventilation

E) Tidal volume

Correct Answer: D. Alveolar Ventilation

Explanation: Alveolar ventilation per minute ($\dot{V}_A$) is specifically the total volume of new, fresh air that reaches the gas exchange regions (respiratory bronchioles, alveolar ducts, alveolar sacs, and alveoli) of the lungs each minute. It is calculated as $\dot{V}_A = \text{Freq} \times (V_T - V_D)$. Pulmonary ventilation (or minute ventilation) represents the total volume of air inspired and expired per minute, including the air that is wasted in the anatomical dead space ($\dot{V}_E = \text{Freq} \times V_T = 6 \text{ L/min}$).

37. Common partial pressure of oxygen in the interstitial fluid of peripheral tissues is

A) 95 mm Hg

B) 20 mm Hg

C) 45 mm Hg

D) 104 mm Hg

E) None of the above

Correct Answer: E. None of the above

Explanation: Under normal, resting physiological conditions, the partial pressure of oxygen (PO_2)

) in the interstitial fluid of peripheral tissues is 40 mm Hg (while cellular

intracellular PO_2 ranges from 5 to 23 mm Hg). Because the value of 40 mm Hg is not listed among choices A, B, C, or D, the correct answer is E. None of the above .

38. Neurons in the brain stem primarily promotes the contraction of diaphragm and external intercostals muscles during inspiration

- A) Apneustic center
- B) Dorsal respiratory group
- C) Nucleus of the nucleus ambiguus
- D) Pneumotaxic center
- E) Ventral respiratory group

Correct Answer: B. Dorsal respiratory group

Explanation: The Dorsal Respiratory Group (DRG) of neurons, located in the nucleus of the tractus solitarius (NTS) in the medulla oblongata, controls the basic rhythm of breathing. Its primary role is to emit rhythmic, repetitive inspiratory neuronal ramp signals that travel down the phrenic and intercostal nerves to cause the contraction of the primary muscles of inspiration (the diaphragm and the external intercostal muscles).

39. Which of the following true about chloride shift

- A) Promotes the transport of oxygen in the blood.
- B) Facilitate HCO_3^- – diffuse into, and Cl^- diffuse out of, red blood cells during the of gases in tissues
- C) Maintains electrical neutrality in red blood cells and the plasma.
- D) Symporters of Cl^- HCO_3^- –

E) All of the above.

Correct Answer: C. Maintains electrical neutrality in red blood cells and the plasma.

Explanation: During gas exchange in systemic tissues, carbon dioxide enters the RBC and is converted into bicarbonate (HCO_3^-) and H^+ . As the bicarbonate concentration rises, HCO_3^- diffuses out of the red blood cell into the plasma down its concentration gradient. To replace these lost negative charges and maintain electrical neutrality, chloride (Cl^-) ions diffuse into the red blood cell from the plasma. This rapid exchange of HCO_3^- and Cl^- in opposite directions is mediated by an anion exchanger carrier protein (an antiporter, not a symporter) and is known as the chloride shift.

40. The amount of air occupying the anatomical dead space

A) 100 mL

B) 150 mL

C) 500 mL

D) 350 mL

E) 1200 mL

Correct Answer: B. 150 mL

Explanation: In a healthy, young adult male of average size, the anatomical dead space (the total internal volume of the non-gas-exchanging conducting zone of the airways, such as the nose, pharynx, trachea, and bronchi) is approximately 150 mL (or roughly 2 mL/kg of body weight).

Dr. Doc:

Exam: Physiology

Year: 2016 E.C

Dr. Doc Tutorial

1. Which nerve provides parasympathetic innervation to the smooth muscles of the bronchi and bronchioles, causing bronchoconstriction?

- A. Vagus nerve
- B. Phrenic nerve
- C. Glossopharyngeal nerve
- D. Trigeminal nerve
- E. Hypoglossal nerve

Correct Answer: A. Vagus nerve

Physiological Explanation: Parasympathetic innervation of the lungs is supplied by the vagus nerve. The preganglionic parasympathetic fibers release acetylcholine (ACh) which acts on muscarinic receptors on bronchial smooth muscle cells, causing mild to moderate bronchoconstriction. In contrast, the phrenic nerve supplies the diaphragm (motor), and the glossopharyngeal nerve transmits sensory signals from the carotid bodies.

2. Activation of histamine receptors on bronchial smooth muscle cells leads to the contraction of the smooth muscle, resulting in:

- A. Increased mucus production
- B. Decreased airway resistance
- C. Increased airflow
- D. Bronchoconstriction

E. Bronchodilation

Correct Answer: D Bronchoconstriction

Explanation: Histamine binds to receptors on bronchial smooth muscle cells, triggering intracellular signaling pathways that lead to muscle contraction and narrowing of the airways

(bronchoconstriction). Histamine and slow-reacting substance of anaphylaxis (leukotrienes) are released in lung tissues by mast cells during allergic reactions, playing a major role in airway obstruction that occurs in allergic asthma.

3. When oxygen moves through the thin side of the blood-gas barrier from the alveolar gas to the hemoglobin of the red blood cell, it traverses the following layers in order?

- A. Epithelial cell, surfactant, interstitium, endothelial cell, plasma, red cell membrane
- B. Surfactant, epithelial cell, interstitium, endothelial cell, plasma, red cell membrane
- C. Surfactant, endothelial cell, interstitium, epithelial cell, plasma, red cell membrane
- D. Epithelium cell, interstitium, endothelial cell, plasma, red cell membrane
- E. Surfactant, epithelial cell, interstitium, endothelial cell, red cell membrane

Correct Answer: B. Surfactant, epithelial cell, interstitium, endothelial cell, plasma, red cell membrane

Physiological Explanation: To diffuse from the alveolar air space into the hemoglobin inside the erythrocyte, oxygen must cross the respiratory membrane layers and blood components in the following anatomical sequence: (1) Surfactant layer lining the alveolus, (2) Alveolar epithelial cell, (3) Interstitial space, (4) Capillary endothelial cell, (5) Blood plasma, and (6) Red blood cell membrane.

4. Ventilation-perfusion matching refers to the matching of:

- A. Amount of air reaching the alveoli with the amount of blood flowing through the pulmonary capillaries
- B. Amount of oxygen in the blood with the amount of carbon dioxide in the alveoli
- C. Rate of breathing with the metabolic demands of the body
- D. Amount of oxygen in the blood with the amount of oxygen in the tissues
- E. None of the above

Correct Answer: A. Amount of air reaching the alveoli with the amount of blood flowing through the pulmonary capillaries

Physiological Explanation: Ventilation-perfusion matching (represented as the

$\dot{V}_A/\dot{Q}$ ratio) is the

precise physiological matching of the rate of alveolar ventilation (the amount of fresh air reaching the alveoli) with the rate of capillary blood flow (perfusion) in the adjacent alveolar capillaries. This ensures optimal blood aeration and gas exchange.

5. A cerebrovascular accident that affects forced expirations during rest and exercise most likely damaged which neural area?

- A. Pneumotaxic center
- B. Apneustic center
- C. Phrenic nerve center
- D. Dorsal respiratory group
- E. Ventral respiratory group

Correct Answer: E. Ventral respiratory group

Physiological Explanation: The Ventral Respiratory Group (VRG) contains expiratory neurons (Eneurons) located in the nucleus retroambiguus caudally. While the VRG is inactive during quiet resting breathing, it becomes highly active during forced expiration and increased ventilatory demand (such as during exercise), supplying active motor signals to the abdominal muscles and internal intercostals.

6. As blood passes through systemic capillaries:

- A. pH rises
- B. HCO_3^- ions pass from plasma to red cells
- C. Cl^- ion concentration in red cells falls
- D. Its oxygen dissociation curve shifts to the left
- E. Its ability to deliver oxygen to the tissues is enhanced

Correct Answer: E. Its ability to deliver oxygen to the tissues is enhanced

Physiological Explanation: In systemic tissue capillaries, metabolizing cells dump CO_2 and hydrogen ions (H^+) into the blood. This decreases blood pH and increases PCO_2

, shifting the oxygenhemoglobin dissociation curve to the right (the Bohr effect). A rightward shift lowers hemoglobin's affinity for oxygen, thereby directly enhancing the delivery of oxygen to the tissues.

7. At a high altitude where atmospheric pressure is halved, there is an increase in:

- A. Pulmonary ventilation
- B. Alveolar H₂O vapor pressure
- C. Arterial PO₂
- D. Arterial pH
- E. None of the above

Correct Answer: A. Pulmonary ventilation

Physiological Explanation: At high altitudes, the decreased barometric pressure lowers the ambient PO₂, resulting in arterial hypoxia (PO₂ < 60 mm Hg). This hypoxia strongly stimulates the peripheral chemoreceptors in the carotid and aortic bodies, triggering a reflex increase in pulmonary ventilation to bring more oxygen into the lungs.

8. Shift of the oxygen dissociation curve of blood to the right:

- A. Occurs in the pulmonary capillaries
- B. Occurs if blood temperature falls
- C. Favours oxygen delivery to the tissues
- D. Favours oxygen uptake from the lungs by alveolar capillary blood
- E. Occurs if 2, 3-DPG decreases

Correct Answer: C. Favours oxygen delivery to the tissues

Physiological Explanation: A rightward shift of the oxygen dissociation curve indicates a reduced binding affinity of hemoglobin for oxygen. This lower affinity means that hemoglobin holds onto oxygen less tightly and releases it more easily, which directly favors and enhances the delivery and unloading of oxygen to the metabolizing tissues.

9. In a measurement of FRC by helium dilution, the original and final helium concentrations were 10% and 6%, and the spirometer volume was kept at 5 liters. What was the volume of the FRC in liters? (Choose the closest option)

- A. 2.5
- B. 3.0
- C. 3.3
- D. 3.8
- E. 5.0

Correct Answer: C. 3.3

Physiological Explanation: Using the conservation of mass equation for helium dilution: $C_1 \times V_1 = C_2 \times (V_1 + \text{FRC})$. Rearranging to solve for FRC: $\text{FRC} = V_1 \times (C_1 - C_2) / C_2$. Plugging in the values: $\text{FRC} = 5 \text{ L} \times (10\% - 6\%) / 6\% = 5 \times 4/6 = 20/6 = 3.33 \text{ L}$.

10. The fall in pulmonary vascular resistance on exercise is caused by:

- A. Decrease in pulmonary arterial pressure
- B. Decrease in pulmonary venous pressure
- C. Increase in alveolar pressure

D. Distension of pulmonary capillaries

E. Alveolar hypoxia

Correct Answer: D. Distension of pulmonary capillaries

Physiological Explanation: During exercise, the marked increase in cardiac output causes a moderate elevation in pulmonary arterial pressure. This pressure rise forces closed capillaries to open (capillary recruitment) and already open capillaries to expand (capillary distension). These two physical mechanisms dramatically lower the total pulmonary vascular resistance, allowing high blood flows without a massive pressure spike.

11. In zone 2 of the lung:

A. Alveolar pressure exceeds arterial pressure

B. Venous pressure exceeds alveolar pressure

C. Venous pressure exceeds arterial pressure

D. Blood flow is determined by arterial pressure minus alveolar pressure

E. Blood flow is unaffected by arterial pressure

Correct Answer: D. Blood flow is determined by arterial pressure minus alveolar pressure

Physiological Explanation: In Zone 2 of the lung (typically located at the apex of an upright individual), the hydrostatic pressure relationships are: Arterial pressure exceeds Alveolar pressure, which in turn exceeds Venous pressure ($P_a > P_A > P_v$). Blood flow is intermittent and depends strictly on the difference between pulmonary arterial pressure (P_a) and alveolar pressure (P_A), rather than the standard arterial-venous pressure difference.

12. The apex of the upright human lung compared with the base has:

- A. A higher PO₂
- B. Normal ventilation-perfusion ratio
- C. A lower pH in end-capillary blood
- D. A higher blood flow
- E. Smaller alveoli

Correct Answer: A. A higher PO₂

Physiological Explanation: In an upright lung, gravity causes blood flow to decrease dramatically from base to apex, while ventilation decreases much more slowly. Consequently, the apex has an abnormally high ventilation-perfusion (

V_A/Q) ratio (up to 2.5 times the ideal value). Because ventilation

is extremely high relative to the blood flow, little oxygen is carried away, resulting in a significantly higher alveolar PO₂

at the apex.

13 . An increase in which of the following increases the O₂ affinity of hemoglobin?

- A. Temperature
- B. PCO₂
- C. H⁺ concentration
- D. 2,3-DPG
- E. Carbon monoxide

Correct Answer: E. Carbon monoxide

Physiological Explanation: Carbon monoxide (CO) binds to hemoglobin with an affinity about 210 times greater than that of oxygen, forming carboxyhemoglobin. When CO binds to one or more of the four heme sites, it causes a conformational change that shifts the oxygen-hemoglobin dissociation curve to the left. This leftward shift represents an increase in the affinity of the remaining hemoglobin binding sites, locking oxygen onto the molecule and preventing its release to tissues.

14. Most of the carbon dioxide transported in the arterial blood is in the form of:

- A. Dissolved
- B. Bicarbonate
- C. Attached to hemoglobin
- D. Carbamino compounds
- E. Carbonic acid

Correct Answer: B. Bicarbonate

Physiological Explanation: Carbon dioxide is transported in arterial blood in three distinct forms:

(1) Dissolved in plasma (7%), (2) As carbamino compounds bound to proteins (23%), and (3) As bicarbonate ions (HCO^-)

3) in the plasma and red cells (70%). Thus, bicarbonate is by far the predominant transport mechanism.

15. Pulmonary surfactant is produced by:

- A. Alveolar macrophages
- B. Goblet cells
- C. Basal cells
- D. Type I alveolar cells
- E. Type II alveolar cells

Correct Answer: E. Type II alveolar cells

Physiological Explanation: Pulmonary surfactant is synthesized, stored, and secreted by Type II alveolar pneumocytes (or surfactant cells). Surfactant is a complex mixture of lipids and proteins (primarily dipalmitoyl phosphatidylcholine, DPPC, or lecithin) that dramatically reduces alveolar surface tension, preventing small alveoli from collapsing.

16. Concerning contraction of the diaphragm:

- A. The nerves that are responsible emerge from the spinal cord at the level of the lower thorax
- B. It tends to flatten the diaphragm
- C. It reduces the lateral distance between the lower rib margins
- D. It causes the anterior abdominal wall to move in
- E. It raises intrapleural pressure

Correct Answer: B. It tends to flatten the diaphragm

Physiological Explanation: The diaphragm is a dome-shaped sheet of skeletal muscle innervated by the phrenic nerves (which originate from spinal cord levels C3, C4, and C5). When it contracts, it flattens and moves downward, which increases the vertical and transverse dimensions of the thoracic cavity, lowering intrapleural and alveolar pressures to draw air in.

17. If the pressures in the capillaries and interstitial space at the top of the lung are 3 and 0 mm Hg, respectively, and the colloid osmotic pressures of the blood and interstitial fluid are 25 and 5 mm Hg, respectively, what is the net pressure in mm Hg moving fluid into the capillaries?

- A. 17
- B. 20
- C. 23
- D. 27
- E. 25

Correct Answer: A. 17

Physiological Explanation: Let's calculate using Starling forces. Outward forces (filtration out of capillaries): Capillary hydrostatic pressure ($P_c = 3$ mm Hg) + Interstitial fluid colloid osmotic pressure ($\pi_i = 5$ mm Hg) = 8 mm Hg. Inward forces (absorption into capillaries): Plasma colloid osmotic pressure ($\pi_c = 25$ mm Hg) + Interstitial hydrostatic pressure ($P_i = 0$ mm Hg) = 25 mm Hg. Net absorption pressure = Inward forces - Outward forces = $25 - 8 = 17$ mm

18. Concerning normal expiration during resting conditions:

- A. Expiration is generated by the expiratory muscles
- B. Alveolar pressure is less than atmospheric pressure

- C. Intrapleural pressure gradually falls during the expiration
- D. Passive due to surface tension in the alveoli and elastic tissue recoil
- E. Diaphragm moves down as expiration proceeds

Correct Answer: D. Passive due to surface tension in the alveoli and elastic tissue recoil

Physiological Explanation: Under normal quiet resting conditions, expiration is a passive process. It does not require any muscle contraction. Instead, it is driven entirely by the relaxation of the inspiratory muscles (primarily the diaphragm) and the passive elastic recoil of the lung tissue combined with the alveolar surface tension force.

19. Concerning the respiratory centers:

- A. The normal rhythmic pattern of breathing originates from neurons in the motor area of the cortex
- B. During quiet breathing, expiratory neurons re actively
- C. Impulses from the pneumotaxic center can stimulate inspiratory activity
- D. The cortex of the brain can override the function of the respiratory centers
- E. The only output from the respiratory centers is via the phrenic nerves

Correct Answer: D. The cortex of the brain can override the function of the respiratory centers

Physiological Explanation: The cerebral cortex of the brain provides a voluntary pathway for controlling respiration, allowing individuals to voluntarily override the brainstem's automatic respiratory centers (e.g., hyperventilating, holding one's breath). Under quiet resting conditions, expiratory neurons do not re actively (quiet expiration is passive).

20. Regarding the measurement of Functional residual Capacity:

- A. Helium dilution does not measure unventilated spaces in the chest
- B. Body plethysmography is inaccurate if used
- C. Helium used to decrease air ow viscosity
- D. All of the above
- E. None of the above

Correct Answer: A. Helium dilution does not measure unventilated spaces in the chest

Physiological Explanation: Helium dilution is a spirometric method that depends on the physical mixing of helium within ventilated lung zones. Therefore, it cannot measure gas trapped in unventilated regions of the lungs (such as behind occluded airways in severe emphysema or asthma). In contrast, body plethysmography is highly accurate and measures all intrathoracic gas volumes, including unventilated areas.

21. A healthy, 45-year-old man is reading the newspaper. Which of the following muscles are used for quiet breathing?

- A. Diaphragm and external intercostals
- B. Diaphragm and internal intercostals
- C. Diaphragm only
- D. Internal intercostals and abdominal recti
- E. Sternocleidomastoid muscles

Correct Answer: A. Diaphragm and external intercostals

Physiological Explanation: Normal quiet breathing utilizes the primary muscles of inspiration: the diaphragm (which contracts to increase the vertical dimension) and the external intercostals (which pull the ribs upward and forward to increase lateral and anteroposterior diameters). Quiet expiration is passive, requiring no muscles.

22. A healthy, 18 years old medical student participates in a 10-km charity run for JUMC. Which of the following muscles does the student use (contract) during expiration?

- A. Diaphragm and external intercostals
- B. Diaphragm and internal intercostals
- C. Diaphragm only
- D. Internal intercostals and abdominal recti
- E. Scaleni and Sternocleidomastoid muscles

Correct Answer: D. Internal intercostals and abdominal recti

Physiological Explanation: During strenuous physical activity like a 10-km run, expiration ceases to be passive and becomes an active process (forced expiration) to speed up lung emptying. This requires active contraction of the expiratory muscles, which are primarily the internal intercostal muscles (to pull the ribs downward) and the abdominal recti (to compress abdominal organs and push the diaphragm upward).

23. A preterm infant has a surfactant deficiency. Without surfactant, many of the alveoli collapse at the end of each expiration, which in turn leads to pulmonary failure. Which of the following sets of changes are present in the preterm infant, compared to a normal infant?

- A. Decreased Alveolar surface tension and decreased Pulmonary compliance
- B. Decreased Alveolar surface tension and increased Pulmonary compliance
- C. Increased Alveolar surface tension and decreased Pulmonary compliance
- D. Increased Alveolar surface tension and No change in Pulmonary compliance
- E. No change in Alveolar surface tension and Pulmonary compliance

Correct Answer: C. Increased Alveolar surface tension and decreased Pulmonary compliance

Physiological Explanation: In a preterm infant with Infant Respiratory Distress Syndrome (IRDS), the deficiency of pulmonary surfactant results in highly elevated alveolar surface tension forces. This high surface tension creates a powerful tendency for the alveoli to collapse, making the lungs very stiff and difficult to expand, which is reflected as a severely decreased pulmonary compliance.

24. A patient has a dead space of 150 ml, functional residual capacity of 3 L, tidal volume of 650 ml, expiratory reserve volume of 1.5 L, total lung capacity of 8 L, and respiratory rate of 15 breaths/min. What is the residual volume?

- A. 500 ml
- B. 1000 ml
- C. 1500 ml
- D. 2500 ml

Correct Answer: C. 1500 ml

Physiological Explanation: Functional Residual Capacity (FRC) is defined as the sum of the

Expiratory Reserve Volume (ERV) and the Residual Volume (RV): $FRC = ERV + RV$. Rearranging the formula to solve for Residual Volume: $RV = FRC - ERV$. Plugging in the values: $RV = 3.0\text{ L} - 1.5\text{ L} = 1.5\text{ L} = 1500\text{ ml}$.

25. From the forces governing the diffusion of a gas through a biological membrane, which of the following changes decreases the diffusion of a gas through a biological membrane?

- A. Increased pressure difference across the membrane (ΔP)
- B. Increased cross-sectional area of the membrane (A)
- C. Increased solubility of the gas (S)
- D. Increased distance of diffusion (d) and the molecular weight of the gas (MW)
- E. All except D

Correct Answer: D. Increased distance of diffusion (d) and the molecular weight of the gas (MW)

Physiological Explanation: According to Fick's Law of Diffusion, the net rate of gas diffusion across the blood-gas barrier is directly proportional to the partial pressure gradient (ΔP), the surface area (A), and the gas solubility (S), and is inversely proportional to the membrane thickness or distance (d) and the square root of the molecular weight (MW). Thus, increasing the distance (d) and molecular weight (MW) decreases diffusion.

26. The basic rhythm of respiration is generated by neurons located in the medulla. Which of the following limits the duration of inspiration and increases respiratory rate?

- A. Apneustic center
- B. Dorsal respiratory group
- C. Nucleus of the tractus solitarius D. Pneumotaxic center
- E. Ventral respiratory group

Correct Answer: D. Pneumotaxic center

Physiological Explanation: The pneumotaxic center, located in the nucleus parabrachialis of the upper pons, controls the 'switch-off' point of the inspiratory ramp signal generated by the Dorsal Respiratory Group (DRG). By limiting the duration of inspiration, it shortens the overall respiratory cycle and secondarily increases the respiratory rate.

27. Obstructive airways disease is similar to restrictive lung disease in that it reduces:

- A. Total lung capacity (TLC) B. The forced expiratory volume in one second (FEV1) C. The ratio FEV1/FVC
- D. Residual volume E. Expiratory flow rate to the same degree

Correct Answer: B. The forced expiratory volume in one second (FEV1)

Physiological Explanation: Both obstructive and restrictive lung diseases lead to a reduction in the volume of air expired in the first second of a forced exhalation (FEV1). However, they differ in their pathophysiology: obstructive disease reduces FEV1 due to high airway resistance (which also severely decreases the FEV1/FVC ratio), whereas restrictive disease reduces FEV1 due to the overall decrease in lung volume (retaining a normal or high FEV1/FVC ratio).

28. Which of the following factors is associated with decreased lung volume?

- A. People living at high altitudes
- B. Taller people
- C. Professional athletes
- D. Heavy smokers
- E. Male sex

Correct Answer: D. Heavy smokers

Physiological Explanation: Heavy smokers suffer from chronic airway irritation and inflammation, which can progress to obstructive lung diseases such as chronic emphysema and bronchiolar obstruction. The destruction of lung parenchyma, chronic infections, and mucosal edema reduce the effective ventilated and functional lung volumes. Male sex, tall stature, and athletic conditioning are associated with larger lung volumes.

29. Which one of the following is correct about Cyanosis?

- A. It may be caused by high levels of carboxyhaemoglobin in the blood
- B. It may be caused by high levels of methaemoglobin in the blood
- C. It is seen in fingers of hands immersed in iced water
- D. It occurs more easily in anaemic than in polycythaemic patients
- E. It is severe in cyanide poisoning

Correct Answer: C. It is seen in fingers of hands immersed in iced water

Physiological Explanation: When hands are immersed in ice water, the intense localized cold triggers protective vasoconstriction of peripheral skin arterioles. Blood flow slows to a crawl, allowing capillaries to extract extra oxygen from hemoglobin. This excessive buildup of deoxygenated hemoglobin ($> 5 \text{ g/dL}$) turns the blood dark blue-purple, which shows through the skin as localized cyanosis. Carbon monoxide poisoning results in cherry-red blood, which masks cyanosis.

30. A healthy 25-year-old has climbed from sea level to a height of 1000 metres, feels fine and is enjoying the view. What is the most appropriate set of arterial blood gas results of this person?

- A. Normal pH, reduced PO_2 , normal PCO_2 , and normal bicarbonate
- B. Normal pH, Normal PO_2 , normal PCO_2 , and normal bicarbonate
- C. Reduced pH, reduced PO_2 , normal PCO_2 , and normal bicarbonate
- D. Increased pH, increased PO_2 , reduced PCO_2 , and reduced bicarbonate
- E. Increased pH, reduced PO_2 , reduced PCO_2 , and reduced bicarbonate

Correct Answer: B. Normal pH, Normal PO_2 , normal PCO_2 , and normal bicarbonate

Physiological Explanation: Climbing to 1000 meters (about 3,300 feet) is a very mild altitude increase. At this level, atmospheric pressure drops slightly (from 760 to about 674 mm Hg), and alveolar PO_2

remains high enough to keep hemoglobin nearly fully saturated ($> 95\%$) on the flat

plateau of the oxygen dissociation curve. Since no tissue hypoxia is present, there is no hyperventilation or compensatory response. Thus, pH, PO_2

, PCO_2 , and bicarbonate levels remain completely normal.

31. Which of the following conditions would lead to a decrease in oxygen content?

- A. Hyperventilation
- B. Anemia
- C. Pulmonary embolism
- D. Increased cardiac output
- E. Hypoxia

Correct Answer: B. Anemia

Physiological Explanation: Anemia is characterized by a decrease in red blood cells or hemoglobin concentration. Because 97% of blood oxygen is carried bound to hemoglobin, a reduction in hemoglobin concentration directly reduces the oxygen-carrying capacity and total oxygen content of arterial blood. This occurs even though the partial pressure of oxygen (PO₂)

) dissolved in the plasma remains completely normal.

32. Which one of the following is wrong about the effects of hydrostatic pressure gradients in the lungs on regional pulmonary blood flow?

- A. The blood pressure in the foot of a standing person can be as much as 90mmHg greater than the pressure at the level of the heart
- B. The pulmonary arterial pressure in the uppermost portion of the lung of a standing person is about 15mmHg lesser than the pulmonary arterial pressure at the level of the heart
- C. The pressure in the upper portion of the lungs is about 8mmHg greater
- D. There is little blood flow in the top (apex) of the lung
- E. There is about 5 times as much flow in the bottom (base) of the lung

Correct Answer: C. The pressure in the upper portion of the lungs is about 8mmHg greater

Physiological Explanation: Option C is incorrect (and thus the correct choice) because the slides state that the pulmonary arterial pressure in the lowest (base) portion of the lungs is about 8 mm Hg greater than at the heart level, while the pressure in the uppermost portion (apex) is about 15 mm Hg less. Therefore, pressure in the upper portion is less, not greater.

33. Which of the following is true about the transport of gases in the blood?

- A. Oxygen is mainly transported in the blood bound to hemoglobin
- B. Carbon dioxide is mainly transported in the blood dissolved in plasma
- C. Oxygen and carbon dioxide are primarily transported in the blood as bicarbonate ions
- D. Oxygen and carbon dioxide are primarily transported in the blood as free gases
- E. Oxygen and carbon dioxide are primarily transported in the blood as hydrogen ions

Correct Answer: A. Oxygen is mainly transported in the blood bound to hemoglobin

Physiological Explanation: Under normal physiological conditions, oxygen is primarily (97%) transported chemically bound to hemoglobin inside the red blood cells, with only 3% dissolved in the plasma. Carbon dioxide is mainly transported as bicarbonate ions (70%), and only 7% is dissolved.

34. Which of the following statements is not accurate regarding zone 3 of pulmonary blood flow?

- A. Blood flow in zone 3 is influenced by the disparity between arterial and alveolar pressures

- B. Blood flow in zone 3 is governed by the typical arterial-venous pressure difference
- C. Zone 3 is localized in the lower regions of the lungs
- D. In zone 3, the alveolar capillary pressure remains higher than the alveolar air pressure throughout the entire cardiac cycle
- E. The pressure gradient in zone 3 is represented as $P_a > P_v > P_A$

Correct Answer: A. Blood flow in zone 3 is influenced by the disparity between arterial and alveolar pressures

Physiological Explanation: In Zone 3 of the lung, the hydrostatic pressures are such that pulmonary capillary pressure remains higher than alveolar air pressure during the entire cardiac cycle ($P_a > P_v > P_A$). As a result, blood flow is continuous and determined strictly by the standard arterial-venous pressure difference ($P_a - P_v$), and is completely unaffected by alveolar pressure (P_A).

35. One of the following is wrong about expiration?

- A. Inspiratory muscles relax & the rib cage descends
- B. The vertical dimension of the chest cavity is increased
- C. Increases in intrapulmonary pressure
- D. Contraction of the abdominal muscles
- E. Expiration is passive during quiet breathing

Correct Answer: B. The vertical dimension of the chest cavity is increased

Physiological Explanation: Option B is incorrect (and thus the correct choice) because during expiration, the vertical dimension of the chest cavity decreases (not increases) as the diaphragm relaxes and is pushed upward by the abdominal viscera, narrowing the thoracic cage.

36. Which of the following factors primarily determines the oxygen content of arterial blood?

- A. Oxygen saturation
- B. Hemoglobin concentration
- C. Partial pressure of oxygen
- D. Carbon dioxide levels
- E. pH levels

Correct Answer: B. Hemoglobin concentration

Physiological Explanation: The oxygen content of arterial blood is calculated as: $\text{Content} = (1.34 \times [\text{Hb}] \times \text{SO}_2$

$/100) + (0.003 \times \text{PO}_2)$. Because 97% of the oxygen is bound to hemoglobin, changes

in the hemoglobin concentration ([Hb]) have the most dominant impact on the total oxygen content of blood.

37. Minute ventilation refers to:

- A. The volume of air inspired or expired in one breath
- B. The total volume of air inspired or expired per minute
- C. The exchange of oxygen and carbon dioxide in the alveoli

- D. The capacity of the lungs to hold air
- E. The maximum volume of air that can be forcibly exhaled after a maximal inspiration

Correct Answer: B. The total volume of air inspired or expired per minute

Physiological Explanation: Minute ventilation (or minute respiratory volume) is the total volume of new air moved into the respiratory passages each minute. It is calculated by multiplying the Tidal Volume (VT) by the breathing frequency per minute (Freq).

38. An increase in respiratory rate coupled with a decrease in tidal volume would result in:

- A. Increased minute ventilation
- B. Decreased minute ventilation
- C. No change in minute ventilation
- D. Increased dead space volume
- E. Decreased oxygen saturation

Correct Answer: C. No change in minute ventilation

Physiological Explanation: Since Minute Ventilation is the direct mathematical product of respiratory rate and tidal volume ($MV = \text{Freq} \times VT$), an equal and opposite change in both parameters (e.g., doubling the rate while halving the volume) will result in no net change in the minute ventilation. Note, however, that alveolar ventilation would decrease because more air is wasted in the anatomical dead space.

39. An increase in dead space volume would result in:

- A. Increased alveolar ventilation rate
- B. Decreased alveolar ventilation rate
- C. No change in alveolar ventilation rate
- D. Increased tidal volume
- E. Decreased respiratory rate

Correct Answer: B. Decreased alveolar ventilation rate

Physiological Explanation: Alveolar ventilation rate (VA) is the actual volume of fresh air reaching the respiratory zone per minute, calculated as: $VA = \text{Freq} \times (VT - VD)$. Therefore, increasing the physiological dead space volume (VD) directly reduces the numerator, leading to a decreased alveolar ventilation rate.

40. Which of the following factors can decrease the respiratory rate?

- A. Increased oxygen levels in the blood
- B. Decreased carbon dioxide levels in the blood
- C. Stimulation of the central chemoreceptors
- D. Activation of the parasympathetic nervous system E. Increased lung compliance

Correct Answer: B. Decreased carbon dioxide levels in the blood

Physiological Explanation: The chemical drive to breathe is primarily dictated by carbon dioxide (PCO_2) and pH acting on central chemoreceptors in the medulla. A decrease in carbon dioxide levels

(hypocapnia) reduces this chemical stimulation, decreasing the firing of the respiratory centers and lowering the respiratory rate.